



MD. ISMAIL HOSEN

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ACADEMIC PROFILE

Computer Science graduate (BSc completed July 2026) with a strong foundation in **backend development, database management, and system design**. Research experience includes **software engineering and AI-assisted software testing**, with a **published paper on supply chain management systems**. Experienced in building scalable applications using **Node.js, Express.js, Spring Boot, and relational databases**. Seeking to pursue postgraduate study in **computer science, software engineering, and related computing disciplines** to advance expertise in **software engineering, backend systems, and AI-driven software architectures**.

EDUCATION

Bachelor of Science in Computer Science — Completed <i>Independent University, Bangladesh</i>	September 2021 - July 2026 <i>Dhaka, Bangladesh</i>
Alim(Higher Secondary Certificate/HSC Equivalent) – Science Group <i>Tamirul Millat Kamil Madrasah</i>	July 2018 – January 2021 <i>Dhaka, Bangladesh</i>
Dakhil(Secondary School Certificate/SSC Equivalent) – Science Group <i>Tamirul Millat Kamil Madrasah</i>	January 2016 – May 2018 <i>Dhaka, Bangladesh</i>

PUBLICATION

Supply Chain Management System Tailored for Agribusinesses 	January 2024 - March 2025
Engineered a comprehensive system design (ERD/Relational Schema) to eliminate intermediaries and streamline farm-to-vendor transactions, enhancing transparency and market access.	
AI-Enhanced Software Testing: A Human-in-the-Loop ... 	September 2025 - March 2026
Designed a scalable, hybrid system architecture for test automation, test case generation, and regression testing, leveraging LLM-based automation while maintaining human-in-the-loop decision-making for complex edge cases.	
A Systematic Approach to Transforming use Cases into Functional 	June 2025 - September 2026
Co-authored an automated LLM-based pipeline for transforming use-case flowcharts into structured functional requirements.	

PROJECTS

ArtBuy  React, Express.js, MongoDB, JWT	September 2024 - January 2025
Developed a full-stack, scalable marketplace web app connecting artists and buyers, featuring secure authentication, inventory management, and structured order workflows.	
AggriCo  React, Express.js, Mysql	January 2024 - April 2024
Architected a full-stack B2B application featuring user-role-based routing and strict product ownership, utilizing unique user ID mapping to ensure 100% data integrity across all operations.	
Bookkit  Backend — Spring Boot, PostgreSQL, JPA	January 2025 - April 2025
Developed a reliable, scalable real-estate web app with layered architecture, strict business logic for data integrity, and DTO-based APIs for maintainability.	
OS Concurrency and Deadlock Resolution  C and Java	September 2025 - October 2025
Implemented solutions for core OS concepts, including Multithreading, CPU Scheduling algorithms (e.g., FCFS, SJF, Round Robin), and Synchronization techniques. Addressed and provided solutions for common concurrency problems such as Deadlock and Starvation.	

TECHNICAL SKILLS

Languages	JavaScript, Java, Python, SQL, HTML, CSS
Frameworks	Express.js, Spring Boot
Databases	PostgreSQL, MySQL, MongoDB
Tools & Other	Git, System Design, Data Structures Algorithms, MS Office Suite

EXPERIENCE

Part-time Teacher (Math & English)	May 2024 - December 2025
<i>Bashundhara Public School and College</i>	<i>Dhaka, Bangladesh</i>
- Taught Mathematics and English to Grades 6-8, independently planning lessons and preparing assignments and examinations. - Managed classroom activities and student learning effectively, demonstrating strong communication, organization, and time-management skills.	

Backend Development Intern	January 2026 - May 2026
<i>Smart Software Ltd</i>	<i>Dhaka, Bangladesh</i>
- Contributed to backend development and RESTful API design using Node.js, Express.js, Spring Boot, and database systems. - Collaborated with senior developers on real-world software projects, gaining hands-on experience in scalable backend architecture and enterprise-level applications.	

KEY ACHIEVEMENTS

- 1st Runner-Up, Robo Soccer – Chittagong Science Carnival 3.0:** Led team strategy and coordination in a robotics competition.
- Team Leadership & Research:** Led a team of 4 to design a **Supply Chain Management System Tailored for Agribusinesses** and **published** associated paper.
- Performance Optimization:** Improved code efficiency for a Line Following Robot (LFR), resulting in a **30% reduction in loading times.**

CERTIFICATIONS

- CS50 - Harvard University
- Career Essentials in Generative AI - Microsoft and LinkedIn
- CCNAv7: Introduction to Networks - Cisco
- Problem Solving (Intermediate) - HackerRank
- SQL (Intermediate) - HackerRank
- Sustainable Summit SDG Brand Champion Awards 2024
- Bangladesh Retail Awards Bangladesh Retail Congress 2026